Misaligned mouse pointer in Kali Linux on QEMU/KVM

Diagnosis and fix for the pointer offset with SPICE and spice-vdagent 0.23 (virt-manager, Xfce)

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TL;DR: spice-vdagent did not refresh the screen geometry after a resolution change in X11 sessions when the package is built with GTK4 (Debian and Kali, from 0.22.1-5 to 0.23.0-3). **It is fixed in spice-vdagent 0.23.0-4**: upgrade and reboot the VM. If you cannot upgrade yet (for example on Ubuntu 26.04 LTS), the vdagent-watch script restarts the agent on every change and keeps the pointer aligned.

1. Symptoms

- The arrow is drawn in one place and the click lands somewhere else: selecting text or hitting menus and buttons becomes nearly impossible.
- At a small resolution that fits entirely inside the viewer window (e.g. 1024×768 or 1280×800) the pointer works fine.
- The offset appears after the resolution changes, especially when raising it to fill the window or the screen (1920×942, 1920×1080).
- Switching the viewer from Wayland to X11, or the video model from Virtio to QXL, does not fix it.

2. Environment where it was reproduced

Component	Version / configuration
Host	Fedora 44 Workstation, GNOME 50 (Wayland), Ryzen 7 7735HS
Virtualization	virt-manager 5.1.0 · QEMU 10.2.2 · libvirt 12.0.0 · system QEMU/KVM connection
Virtual machine	Q35, UEFI without Secure Boot (OVMF_CODE_4M), virtio disk and NIC, SPICE, USB tablet, spice and qemu-ga channels
Guest	Kali Linux 2026.2 (rolling), Xfce on X11
Agent	spice-vdagent 0.23.0-3 (affected version)

3. Cause

When the SPICE agent is running, the pointer works in *client* mode: virt-manager sends the **absolute** mouse position and the agent injects it into Xorg through a virtual device that shows up in `xinput list` as `spice vdagent tablet`. To translate that position, the agent needs to know the screen geometry.

The agent reads the screen size at startup and is supposed to refresh it on every RandR event. In `src/vdagent/x11-randr.c` that refresh was wrapped in `#ifndef USE_GTK_FOR_MONITORS`. Since 0.22.1-5, Debian builds the package with `--with-gtk4=yes`, which defines that macro and leaves the refresh out of the binary. GTK's monitor API only answers on Wayland, so on X11 the agent keeps the startup geometry, the virtual tablet keeps that size and absolute pointer events drift. Restarting the agent makes it read the geometry again.

It is not new code in 0.23.0: `x11-randr.c` is identical in 0.22.1 and 0.23.0; the GTK4 build exposed it. The upstream fix (MR !58) removes those guards, and Debian ships it in **0.23.0-4**, released on October 1, 2026.

Package	Status
Debian stable 0.22.1 (GTK3 build)	Not affected
Debian and Kali GTK4 builds before 0.23.0-4 (e.g. 0.23.0-3)	Affected
Ubuntu 26.04 LTS, 0.23.0-1ubuntu0.1	Affected: GTK4 build without the patch (LP #2168964)
Debian and Kali 0.23.0-4 or later	**Fixed** (verified on Kali 2026.2)

4. How it was diagnosed

Each test ruled out one hypothesis. The same method works for similar problems:

Test	Result	Conclusion
Small resolution, viewer scaling set to “Never”	Pointer OK	The offset depends on the resolution
virt-manager forced to X11 (`GDK_BACKEND=x11`)	Still offset	Not the viewer on Wayland
Video model Virtio → QXL	Still offset	Not the video model
`xinput list`	QEMU USB Tablet and spice vdagent tablet present	Absolute pointer is configured correctly
Restart the session agent at 1920×1080	**Pointer OK**	**It is the agent** (bug #9653)
Compare 0.22.1 and 0.23.0 source	x11-randr.c identical; refresh compiled out with GTK4	Root cause: the GTK4 build
Upgrade to 0.23.0-4 and reboot, no script	**Pointer OK**	**Official fix verified**

5. Immediate fix

If the pointer drifts, restart your session agent from a Kali terminal (keyboard only; no sudo needed):

```
pkill -x spice-vdagent
spice-vdagent
```

The agent goes back to the background on its own and the pointer stays aligned until the next resolution change.

6. Permanent fix: the vdagent-watch script

The script listens for X screen-change events (RandR) with xev and restarts the agent whenever the resolution changes. It does not poll: it sleeps and uses no CPU or battery while nothing happens. On start it restarts the agent once, in case the pointer is already misaligned. It coalesces the burst of events from a single change, avoids restarting twice for the same resolution, logs every action to the journal, and notifies you when spice-vdagent is upgraded so you can check whether the bug is already fixed.

Install

Inside Kali, from a terminal in the graphical session, as your normal user (not root). The installer adds missing packages (x11-utils, x11-xserver-utils), copies the script to `~/.local/bin`, creates the autostart entry and starts it:

```
git clone https://github.com/JorgeXern/kali-spice-vdagent-pointer-fix.git
cd kali-spice-vdagent-pointer-fix
./install.sh
```

Manual install (without git)

Download the `vdagent-watch` file from the repository and run:

```
sudo apt install -y x11-utils x11-xserver-utils
install -Dm755 vdagent-watch ~/.local/bin/vdagent-watch
mkdir -p ~/.config/autostart

cat > ~/.config/autostart/vdagent-watch.desktop <<EOF
[Desktop Entry]
Type=Application
Name=vdagent-watch
Comment=Workaround for misaligned pointer (spice-vdagent 0.23, Kali bug #9653)
Exec=$HOME/.local/bin/vdagent-watch
Terminal=false
EOF
```

Then log out and back in, or start it right away with `~/.local/bin/vdagent-watch &`.

Check that it works

Follow the log, then maximize, restore and go fullscreen: each change should add a line such as `resolution 1920x1080: agent restarted`, and the pointer should line up within a second or two.

```
journalctl -t vdagent-watch -f
pgrep -af vdagent-watch      # two lines = one instance (main + loop)
```

The commented script

Use the file from the repository instead of copying from this PDF: copying can alter spaces or quotes.

```
#!/usr/bin/env bash
#
# vdagent-watch: keep the SPICE mouse pointer aligned in Kali Linux VMs.
#
# Problem  GTK4 builds of spice-vdagent before Debian 0.23.0-4 (e.g. Kali
#          0.23.0-3, Ubuntu 26.04 0.23.0-1ubuntu0.1) keep using the old screen
#          geometry after the guest resolution changes on X11, so the arrow
#          you see and the point you click drift apart. Kali bug #9653:
#          https://bugs.kali.org/view.php?id=9653
#          Fixed in Debian/Kali 0.23.0-4: upgrade and reboot if you can.
#
# Fix      Restarting the per-session agent makes it read the geometry again.
#          This script does that automatically and only when needed: it waits
#          for X RandR events (no polling, no CPU use while idle) and restarts
#          the agent once per real resolution change, plus once at startup in
#          case the pointer is already misaligned.
#
# Remove   once Kali ships a fixed spice-vdagent. The script logs and shows a
#          notification when it finds a version newer than the buggy one.
#
# Needs    xev (x11-utils), xrandr (x11-xserver-utils), spice-vdagent
# Logs     journalctl -t vdagent-watch
# Project  https://github.com/JorgeXern/kali-spice-vdagent-pointer-fix (MIT)

# Fail on unset variables. No `set -e`: grep, pkill and read return non-zero
# in normal operation, and this script must keep running.
set -u
```

```
readonly NAME="vdagent-watch"
readonly BUGGY_VERSION="0.23.0-3"   # last version known to have the bug

log() { logger -t "$NAME" -- "$*"; }
die() { echo "$NAME: $*" >&2; log "$*"; exit 1; }

# --- Preconditions ------------------------------------------------------

for cmd in xev xrandr spice-vdagent flock; do
  command -v "$cmd" >/dev/null || die "missing command '$cmd'"
done
[ -n "${DISPLAY:-}" ] || die "no X display; run it from the graphical session"

# One instance per user: autostart plus a manual run must not race each other.
# The lock lives on fd 9 and is released automatically when the script exits.
exec 9>"${XDG_RUNTIME_DIR:-/tmp}/$NAME-$UID.lock"
flock -n 9 || { echo "$NAME: already running" >&2; exit 0; }

# --- Has Kali shipped a newer agent? ------------------------------------
# A version newer than the buggy one may already contain the fix: say so in
# the journal and, once the desktop is up, as a notification.

version=$(dpkg-query -W -f='${Version}' spice-vdagent 2>/dev/null)
if [ -n "$version" ] &&
   dpkg --compare-versions "$version" gt "$BUGGY_VERSION"; then
  msg="spice-vdagent is now $version. If the pointer stays aligned without"
  msg+=" $NAME, the bug is fixed and you can uninstall it."
  log "$msg"
  if command -v notify-send >/dev/null; then
    # Wait a bit so the notification daemon is up at login.
    { sleep 10; notify-send "$NAME" "$msg"; } &
  fi
fi

# --- Helpers ------------------------------------------------------------

# Print the resolution of the first connected output, e.g. "1920x1080".
# Parses the xrandr line "Virtual-1 connected primary 1920x1080+0+0 (...)".
current_resolution() {
  xrandr --current 2>/dev/null |
    awk '/ connected/ && match($0, /[0-9]+x[0-9]+[+]/) {
           print substr($0, RSTART, RLENGTH - 1); exit }'
}

# Replace this session's agent with a fresh one that reads the new geometry.
# -x matches the exact process name, so the system daemon (spice-vdagentd,
# run by root) is never touched; -u limits the match to our own processes.
# The agent forks to the background by itself. `9>&-` keeps the lock fd out
# of it, otherwise the agent would hold the lock after this script exits.
restart_agent() {
  pkill -x -u "$UID" spice-vdagent
  spice-vdagent 9>&-
}

# --- Main loop ----------------------------------------------------------
# xev prints one block per event; only the header line of each RandR screen
# change matters. --line-buffered makes grep pass each line on immediately
# instead of waiting for its output buffer to fill. `read` blocks without
# using CPU until the next line arrives. The pipeline ends with the X session.
#
# The pipeline runs in the background and the script waits for it, so that
# stopping the script (Ctrl+C, pkill) also stops xev and grep through the
# EXIT trap. Bash would delay the trap while a foreground pipeline runs.

trap 'pkill -P $$' EXIT
trap 'exit 0' INT TERM HUP
```

```
log "started (spice-vdagent ${version:-version unknown})"

xev -root -event randr |
  grep --line-buffered '^RRScreenChangeNotify' |
  {
    # Startup sync: the pointer may already be misaligned when this script
    # starts (the resolution changed before installing it, or early in the
    # login). Give the session's autostart entries a moment, then restart a
    # running agent once. Events that arrive meanwhile wait in the pipe.
    sleep 2
    last=$(current_resolution)
    if pgrep -x -u "$UID" spice-vdagent >/dev/null; then
      restart_agent
      log "startup sync (${last:-unknown resolution}): agent restarted"
    fi

    while read -r _; do
      # A single resize produces a burst of events. Drain everything that
      # arrives within one second, then act once on the final resolution.
      while read -r -t 1 _; do :; done

      current=$(current_resolution)

      # Nothing to do if xrandr returned nothing or the size did not change.
      if [ -z "$current" ] || [ "$current" = "$last" ]; then
        continue
      fi

      restart_agent
      log "resolution $current: agent restarted"
      last=$current
    done
  } &
wait $!
```

7. Official fix: upgrade to 0.23.0-4

This is the recommended solution. Upgrade the package and **reboot the VM**: the running agent keeps using the old binary until it restarts. Until then, `ls -l /proc/$(pgrep -xo spice-vdagent)/exe` ends in "(deleted)".

```
sudo apt update && sudo apt install spice-vdagent
sudo reboot
zcat /usr/share/doc/spice-vdagent/changelog.Debian.gz | head -10   # should cite LP: #2168964
```

Verified on Kali 2026.2: with 0.23.0-4 and without the script, the pointer stays aligned across xrandr changes and viewer window resizes. The script is still useful where the fixed package is not available yet.

8. Configuration recommendations

- In the virt-manager viewer: **View → Scale Display → "Never"**, so the guest is shown 1:1. Then pick one option: **"Auto resize VM with window"** enabled (the resolution follows the window) or a fixed resolution with that option disabled.
- Inside Kali, install the agents: `sudo apt install -y spice-vdagent qemu-guest-agent`.
- Make sure the USB tablet exists: `xinput list | grep -i tablet`.
- If you set a fixed resolution in Xfce's display settings, the script still realigns the pointer after the change. To go back to automatic resizing: `xfconf-query -c displays -p /Default -r -R`.
- Take a VM snapshot before changes like these (View → Snapshots).

9. Removing the workaround

With spice-vdagent 0.23.0-4 or later (the script notifies you when it detects it), upgrade, reboot the VM, confirm the pointer stays aligned without the script and uninstall it:

```
./uninstall.sh      # from the repository folder
```

References

- Kali Linux Bug Tracker #9653: "spice-vdagent, possible regression, mouse pointer misaligns with cursor". https://bugs.kali.org/view.php?id=9653
- Ubuntu Discourse: "Mouse pointer desync/offset on fresh Lubuntu 26.04 (SPICE + VirtIO) after monitor size change". https://discourse.ubuntu.com/t/82365
- spice-vdagent upstream, MR !58 "x11-randr: keep cache current on X11+GTK4 builds". https://gitlab.freedesktop.org/spice/linux/vd_agent/-/merge_requests/58
- Launchpad #2168964 (Ubuntu). https://bugs.launchpad.net/bugs/2168964
- Debian Package Tracker: spice-vdagent (0.23.0-4). https://tracker.debian.org/pkg/spice-vdagent
- Repository for this fix (script, installer and this guide): https://github.com/JorgeXern/kali-spice-vdagent-pointer-fix

The official fix is spice-vdagent 0.23.0-4. Use this workaround only on systems that cannot install it yet.